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Course : Computer Vision

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ACTIVITY

ANALYTICAL PROBLEM-SOLVING

1) Gradient magnitude and Direction

Given:

$$G_x = 24, G_y = 18$$

Gradient Magnitude:

$$G = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{24^2 + 18^2}$$

$$= \sqrt{576 + 324} = \sqrt{900}$$

$$\boxed{G = 30}$$

Gradient Direction:

$$\theta = \tan^{-1}(G_y/G_x)$$

$$= \tan^{-1}(18/24)$$

$$\boxed{\theta \approx 36.87^\circ}$$

2) Non-Maximum Suppression

Given:

Pixel	P1	P2	P3	P4	P5
Magnitude	25	50	90	60	30

Gradient + direction is horizontal, so each pixel is compared with its left + and right neighbours

→ $P_2 = 50$: neighbours 25 and 90 → 50 is smaller
→ suppressed.

→ $P_3 = 90$: neighbours 50 and 60 → 90 is smaller → kept.

→ $P_4 = 60$: neighbours 90 and 30 → 60 is smaller, suppressed.

So,

$P_3 = 90$ is retained.

Other pixels are suppressed at the boundaries after comparison.

3. Double Thresholding

Given:

High Threshold: 100

Low Threshold: 50

Pixel	Magnitude	Classification
P_1	135	strong edge
P_2	90	weak edge
P_3	45	Non-edge
P_4	120	strong edge
P_5	70	weak edge
P_6	20	Non-edge

Therefore:

Strong edges: P_1, P_4

Weak edges: P_2, P_5

Non-edges: P_3, P_6

Edge Tracking by Hysteresis

Given:

Pixel	P1	P2	P3	P4	P5	P6	P7
Magnitude	130	95	40	115	65	25	105

High Threshold: 100

Low Threshold: 50

Strong edges: Magnitude ≥ 100

P1, P4, P7

Weak edges: Magnitude between 50 and 99

P2, P5

Non-edges: Magnitude < 50

P3, P6

Given:

→ P2 is connected to P1 → P7 becomes an edge.

→ P5 is not connected to any strong edge

→ P5 is removed